

Second Meeting – Advanced Systems Lab

Team 80

Meeting Information

Course: Advanced Systems Lab

Team: Team 80

Meeting with: Mohamed-Hicham Leghettas

Notes: Eli

Feedback About the Presentation

- 1) The validation error is a bit high. The mean is acceptable, especially when using Mean Absolute Percentage Error (MAPE):

$$\text{MAPE} = \frac{|\text{predicted value} - \text{actual value}|}{|\text{actual value}|}$$

For values in the millions, an error below 1% is considered good.

- 2) Use the following timing setup:
 - Warmup runs: 10
 - Measured runs: 30
 - The full benchmark with 40 total runs should finish within 2 hours.
- 3) For runtime experiments, varying T is valid. It is also acceptable to vary all parameters at once. However, the suggested setup is:
 - Keep the batch size fixed.
 - Use $C = 2$, $T = 64$, and $L = 16$.
 - Multiply everything by N .
 - Avoid producing multiple separate plots where only T or only L is varied.
- 4) A roofline plot should be produced. The PyTorch reference implementation should be pinned to one CPU core by disabling multithreading, and the C/C++ implementation should appear as a separate point.
- 5) Before implementing optimizations, think carefully about sparsity. CSR is probably not the right choice because the structure is very deterministic. Instead, consider hardcoding multiplications and sign flips.

- 6) After memory layout optimizations, produce another line or plot specifically for that version, since the number of operations may change dramatically.
- 7) A roofline plot is less informative if both pre-optimization and post-optimization versions are shown together. Runtime is likely the most informative metric, so use the most appropriate timing setup.
- 8) Do not switch between AMD and Mac architectures, especially when discussing vectorization.
- 9) Pick one architecture, because the report should focus on only one. If Mac is chosen, then all profiling and timing experiments need to be rerun on the Macs.
- 10) For geometric and equilinear operations, definitely exploit sparsity and memory layout before trying other optimizations.
- 11) Timing plots should use seconds as the unit.